

DTSA 5841

IBM Capstone Project

FORMAT

Online (DTSA)

CATEGORY

Elective

CREDITS

1

SECTION

Section 4 – Elective

BEST ROLE MATCH

Data Scientist (82.4 / 100)

AVG ALIGNMENT

72.4 / 100

SPECIALIZATION Applied Capstone

72CURRENT AVG SCORE
across 6 job roles**95**POTENTIAL SCORE
with recommended additions**+28**MAX SCORE GAIN
best single-role improvement**CURRENT COURSE DESCRIPTION**

The IBM Capstone Project course will allow you to apply the knowledge and skills from the MS-DS degree to a real-world data set provided by IBM. This project will allow you to work independently on a data set that will test your skills in acquiring, cleaning, modeling data, and analyzing a dataset using data mining and machine learning techniques. By the end of this course, you will have a project that you can add to your data science portfolio to show off to employers and demonstrate your data science expertise. This course uses the IBM dataset from the IBM Applied Data Science Capstone course, part of the IBM Data Science Professional Certificate, and provides additional instruction and assessments in order to apply this capstone as an elective for the MS-DS Coursera degree at the University of Colorado Boulder. Because CU is collaborating with IBM on this course, all students will have full access to the IBM Applied Data Science Capstone Course while taking DTSA 5841. It is strongly recommended that you take the Capstone Project course as one of your final courses in the program as you will work with real-world data sets that will use MS-DS core concepts. In order to be successful in this course, only students who have completed the following specializations should register for the Capstone Project: Data Science Foundations: Data Structures and Algorithms, Data Science Foundations: Statistical Inference, Data Mining Foundations and Practice and Machine Learning.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	82.4	100.0	+17.6	MEDIUM
AI Engineer	79.9	100.0	+20.1	MEDIUM
Machine Learning Engineer	79.6	98.8	+19.2	MEDIUM
ML Engineer	76.3	100.0	+23.7	MEDIUM
Applied Scientist	66.3	92.5	+26.2	MEDIUM
MLOps Engineer	50.1	77.9	+27.9	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 5 of 6 roles

systems

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 LOW Optional: Feature Engineering — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

Data Scientist

82.4 → 100.0 (+17.6 pts)

toolkit

presentation

different

problems

AI Engineer

79.9 → 100.0 (+20.1 pts)

models

technical

systems

teams

problems

detection

design

tools

Machine Learning Engineer

79.6 → 98.8 (+19.2 pts)

tools

design

models

applications

technical

systems

analysis

support

ML Engineer

76.3 → 100.0 (+23.7 pts)

systems

models

world

problems

technical

building

tools

detection

Applied Scientist

66.3 → 92.5 (+26.2 pts)

- computational
- research
- literature
- problems
- models
- technical
- systems
- industry

MLOps Engineer

50.1 → 77.9 (+27.9 pts)

- detection
- scientists
- systems
- preprocessing
- tools
- models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.